DATA ANALYTICS RESEARCH ASSIGNMENT 1 — SECTION A

① A database is a systematic approach to data storage and organization that facilitates easy access, management, and updating.

- Databases are essential because ;

1. They aid in accessing information efficiently and effectively as well as storing it which is absolutely imperative in the modern world where people deal with large amounts of content upon limited storage.

- EXAMPLES :

1. Databases are used in hospitals to store and organize and retrieve patient information.
2. They are also used in Telecommunications companies, for instance, MASCOM, VODACOM, CELL C, ORANGE etc to store customer information as well as cell phone numbers.

② Data Warehouse — A single location created especially to hold massive amounts of structured data from many sources for reporting and analysis.

- DATA WAREHOUSE VS. DATA BASE

A data warehouse is meant for analytics which is the Online Analytical Processing or commonly referred to as OLAP which emphasizes data consistency across sources, query efficiency, and historical data storage. Whereas, a database focuses on Online Transaction Processing (OLTP) which aids in daily activities and emphasizes quick transactional data inputs, updates and ensuring the transactional data should need be.

carefully chosen portions of data designed for specific reporting and analytic requirements.

- It relates to a data warehouse in the sense that a data mart is often built from a data warehouse, but it can also exist independently.

- It is ofterly used by specific businesses not engineers or data scientists.

(5) Structured data – Information arranged according to a predetermined schema, usually kept in relational database as rows and columns. Eg: Account numbers.

Semi-structured data – Information that has certain organizational characteristics, such as tags/keys, but does not fit into a conventional table structure. Eg: XML documents, JSON files.

Unstructured data – This is data that is frequently kept in its original format and lacks a predetermined model/framework. Eg: Emails, PDFs, movies and images.

(6) Data modeling is the process of putting data structures and their connections inside a system into a logical/visual representation. It usually uses schemas/diagrams to specify how data is linked, stored and retrieved.

- IMPORTANCE:
It enhances database design by ensuring databases are scalable, efficient and standardized and furthermore

- USES :

1. Financial institutions may need a data warehouse for forecasting, trend analysis and strategic planning.
2. Research institutions like NASA or the World Health Organization need a Datawarehouse for the optimization of complicated searches, aggregations, and read-intensive processes due to the fact that they deal with a vast majority of information with multiple wording hence the datawarehouse to cut in simplicity.

③ Data Lake is where you may store raw, unprocessed data in any format — structured, semi-structured, or unstructured in a data lake, which is a centralized storage system. It is intended for sophisticated data processing, machine learning, and big data analytics.

DIFFERENCES FROM A DATA WAREHOUSE :

- Data lake uses schema-on-read, supports all data types, is ideal for data exploration and sophisticated analytics and stores raw data in its original format whereas a datawarehouse utilizes schema-on-write (data is structured before storage) and store processed, structured data that is ideal for business intelligence and reporting.

- A company would prefer data lake over a data warehouse when it collects massive amounts of data for example, spotify.

④ Data Mart — This is a subset of a data warehouse that concentrates on a particular team, department or business function within a company. It includes a

security, automatic tuning and scalability for cloud-based applications. Ideal for transactional workloads and enterprise applications.

2. ORACLE DATABASE
A popular enterprise-grade relational database system that is renowned for its strength, scalability and sophisticated features including robust security and Real Application Clusters (RAC). It is frequently used in mission-critical systems and large-scale corporate applications.

3. MySQL
A popular open-source relational database management system for small and medium-sized operations and online applications. It is lightweight, simple to use and extensively supported by cloud platforms and hosting companies.

4. Azure Database for PostgreSQL
An open-source, fully managed PostgreSQL database service with AI-ready capabilities, high availability and enterprise level security. In addition to integrating with Azure AI services for sophisticated analytics and intelligent applications, it handles mission-critical workloads.

(9) ETL (Extract, Transform, Load) is whereby data from several sources is combined into a single data repository for analytics and reporting using the Extraction, Transform, Load (ETL) data integration process.

- STEPS:

1. Extract – Information is gathered from a variety of sources, including files, databases and APIs. This stage guarantees the collection of raw data for processing.

2. Transform – In accordance with business requirements, the extracted data is cleaned, verified and reshaped. Filtering, sorting, aggregating, joining, deduplicating and applying computations are examples of common operations. Data consistency and quality are guaranteed by this procedure.

3. Load – To enable querying and analysis, the converted data is imported into a destination system, such as a data lake / warehouse.

- IMPORTANCE

It is imperative because it ensures compliance with regards to data governance and auditability during transformation and loading. Furthermore it supports in data consolidation as it combines data from multiple sources. Other important factors improve efficiency as well as data quality.

(10) Comma – Seperated Values (CSV)

Description: A plain text format with commas between fields and each line denoting a record.

When to use: Not effective for big datasets or intricate structures, but ideal for straight forward tabular data that is simple to understand and distribute.

JSON (JavaScript Object Notation)
Description : Often used for nested / hierarchial data, this text - based format represents structured data as key - value pairs.

When to use : Perfect for situations where human readability and flexibility are crucial, semi - structured data and API's

Parquet
Description : An analytics - focused columnar storage format that facilitates effective encoding and compression

When to use : Ideal for huge dataset searches and big analytics, particularly when with Spark / Azure Synapse.

Avro
Description : Compact and serializable, this row based binary format has schema.

When to use : Frequently utilized in situation requiring schema evolution and effective serialization, as well as in streaming and messaging systems (such as Kafka).

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② https://azure.microsoft.com/en-us/products/synapse-analytics/

data modeling improves performance by efficiently organizing data, searches and storages are optimized.

EXAMPLES : Key - value model, graph model, time series model etc.

② Data Mining — The practice of employing algorithms and statistical approaches to extract patterns, correlations and insights from massive data bases.

USES :

- In businesses it is used for customer insights meaning it facilitates better customer retention, targeted marketing and customisations for instance.

EXAMPLES :

1. SHEIN suggesting clothes I could buy based off what I have bought before.
2. Instagram can suggest content I could like on my "For you page" based off tracking other apps.

③ Database Management System — A software layer that makes it possible for apps and users to communicate with databases. It oversees storage, organization, retrieval and the security of the data, guaranteeing dependable and effective operations.

FOUR POPULAR DBMS PLATFORMS :

1. Azure SQL Database

Microsoft SQL server is the foundation of this fully managed relational database solution. It offers high availability, built-in

problem and in the contrary to machine learning which shows the computer examples so that it learns how to solve the problem.

- SUPERVISED LEARNING : A model is trained to anticipate the correct output for a given input. It is applicable to tasks like regression and classification that call for a certain level of accuracy in relation to an external ground truth.

UNSUPERVISED LEARNING : A model is trained to identify inherent patterns, relationships and correlations in data through unsupervised learning. Unlike supervised learning, unsupervised learning tasks lack an external ground truth to compare their results against.

REINFORCEMENT LEARNING : Here a model is trained to assess its surroundings and choose the ~~ch~~ course of action that will yield the highest reward. While there isn't a single ground truth in RL events, the are good, and bad or neutral acts.

(13) Deep Learning is a branch of machine learning that use neural networks to extract patterns from massive data sets. These models, which are sometimes refferred to as deep neural networks (DNN), are somewhat influenced by biological brain systems. Rather than depending on rules designed for particular activities, they automatically acquire data representations.

- They relate because AI is the umbrella term for everything as machine learning is a subset of AI where systems learn from data instead of being explicitly programmed. Deep learning is a subset of machine learning that uses neural networks with many layers.

classification, and sentiment analysis. Large language models in NLP are trained to produce text or make predictions using patterns and traits taken from input data.

- NLP APPLICATIONS :

1. Email filters on Gmail - This enables one to categorize mails according to their importance instead of everything being mixed up in one place, especially ads or promotions which may even distort important information from being seen.

2. Smart Assistant like Siri - Convenience at its best allowing you to handle multiple tasks at once; make a call while typing or cooking in an instant without necessarily pressing the phone.

3. Search results on Google for instance which makes finding information a lot less excavating and going through a bunch of information before you get to what you want.

(15) Large Language Model : Are a class of deep learning models that have been trained on enormous volumes of data, enabling them to comprehend and produce natural language and other kinds of material to carry out a variety of tasks. LLMs are based on a kind of neural network ~~and~~ architecture known as a transformer, which is particularly good at managing word sequences and identifying textual patterns.

LLM VS STANDARD MACHINE LEARNING :

- Classification, regression, clustering and forecasting are all a vast area of machine learning (ML), whereas on

③ https://azure.microsoft.com/en-us/products/synapse-analytics.

SECTION B: ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

⑪ ARTIFICIAL INTELLIGENCE (AI) – The capacity to gather, process, generate and use knowledge in the form of a model to forecast, suggest or make judgements.

NARROW AI VS GENERAL AI

Narrow/weak AI is a form of artificial intelligence which is trained to complete a particular or specific activity, frequently far more quickly and effectively than human intellect. It is unable to perform functions outside of its assigned duty rather, it focuses on advancements within a certain subset of cognitive talents. Eg: Alexa from Amazon and Siri from Apple. On the contrary, General/Strong AI is AI that functions without the need for humans to educate the underlying models, prior knowledge and abilities maybe employed to complete new tasks in a different environment. Eg:

⑫ Machine Learning – The branch of Artificial Intelligence (AI) which focuses on algorithms than can learn the patterns in training data and then draw reliable conclusions about the new data. Machine learning models may generate predictions or judgements without explicit, hard-coded instructions because to their pattern recognition capability.

- Traditional programming tells the computer how to solve the

THREE TOOLS THAT USE GENERATIVE AI :

1. Speech-to-text : Use domain specific models to accurately translate voice into text in order to enhance quality
2. Video intelligence : In video, identify shots, faces, brands, texts, and more
3. Translation Advanced : Translation support for romanized text, custom models, batch text and structured document

(7) A training dataset is information which aids in training a machine learning model to make predictions, identify patterns, or produce content.

IMPORTANCE OF QUALITY IN TRAINING DATA :

1. Increased Accuracy – Models may generate more accurate predictions and choices when they are trained on reliable and representative data
2. Decreased Bias – The biases that frequently occur when models are trained on homogeneous data sources are lessened by using diverse datasets
3. Enhanced Generalization – AI systems can function more effectively in real-world applications when they are exposed to a range of scenarios and languages
4. Innovation Catalyst – New insights and unique data from various locations might stimulate creative use cases and applications

DANGERS OF BIASED TRAINING DATA

- AI models that are trained on biased data – such as historical or representational bias – may provide skewed or biased results that unfairly deceit or otherwise discriminate against particular groups / people

(18) Computer Vision – Enables visual data from pictures and videos to be interpreted and analyzed by computers. This area of artificial intelligence employs machine learning, deep learning and pattern recognition to recognize objects, find patterns and derive significant insights.

- Similar to how the human eye and brain process and interpret visual input, Computer vision enables robots to do the same. Camera's, sensors and sophisticated algorithms that have been trained on enormous volumes of visual data and pictures are used in computer vision applications.

Core steps in image analysis : Capture the image → interpret the image → analyze and make sense of the data and then deliver insights.

- EXAMPLES
- Facial ~~reg~~ recognition which is used as a security measure to unlock your phone.
- In biometrics, facial recognition maybe used and in corperate companies they maybe used as a security measure in rooms with limited access.

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SECTION C : APIs, TOOLS AND DEVELOPER CONCEPTS

(19) API (Application Programming Interface) – Is a collection of guidlines, procedures and resources that facilitate communication between various software programs.

- APIs are important because they let different programms talk to each other. Instead of building everything from scratch, apps can use APIs to get services/data from other systems. For example, an app can use Google Maps for location / Stripe for payments.

(20) API Key – A code that identifies and verifies a person/application. Platforms like an internal marketplace with a white label can provide API keys. Aditionally, the offer a secret token for authentication and serve as a unique identity.

PURPOSE – To verify and recognize a user / application.

- API keys should never be directly inserted into source files or applications. This information can be readily found and misused if it ends up in a public setting.

the contrary the specialist field of LLMs is devoted to comprehending and producing human language.

EXAMPLES OF LLMS

1. Language Translation - To access a variety of markets, LLMs can translate materials and webpages into many languages
2. Security - To detect threat patterns and automate actions, LLMs may be applied in cybersecurity eg: antivirus

(ii) GENERATIVE AI : A category of AI algorithms that produces original content in response to a users prompt or request.

HOW DOES IT WORK :

- Data is required to train generative AI models. In order for the neural network to learn to identify patterns, it must be exposed to big datasets. Supervised and Unsupervised learning are the two forms of learning that make up the training process. In unsupervised learning, the model investigates the data without explicit labels, discovering intrinsic patterns or groups on its own, whereas in supervised learning, the model is given labelled data, enabling it to learn to train outputs for given inputs.

The model refines its capacity to produce logical and pertinent outputs as it learns to make better predictions repeatedly throughout training. This is accomplished using a procedure known as backpropagation, in which the model uses the input data to create predictions, compares those forecasts to the actual outcomes and then computes an error or loss. The network may gradually improve its capacity to provide accurate and coherent content by fine-tuning its parameters

Highly tied to the quality of the input.

- Tell me about AI (Bad prompt)
- Elaborate more on the dynamics of Artificial Intelligence in the current world. (Good prompt)

- Write a report on digital marketing (Bad prompt)
- With examples of visual representation (bar graphs, pie charts, line graphs) write a detailed report on the fluctuations of the digital market.

(24) MAP function is a higher-order function in various programming languages that applies a function to each member in a collection, such as a list or set, and returns the results in the collection of the same type.

Example : ()

(25) Cloud computing is the process of storing, managing and processing data via an internet-hosted network of distant servers as opposed to a local server or personal computer.

- Infrastructure as a Service - Provides on-demand access to network, server and storage. It Foundation is virtualization, which transforms actual hardware into adaptable resources that can be scaled up or down as required. IaaS allows you to control the data, apps and operating systems while the supplier takes care of the actual infrastructure. With this configuration, you have the most liberty and control, thus making it simple to create and expand your IT environment to your desired size. Eg: Host, Store, Manage.

- Platform as a service – Offers a ready-to-use environment for creating, evaluating and implementing applications. The provider takes care of all the infrastructure and runtime aspects, so you can focus on creating codes and managing data. Eg: Build, automate, test and stage.

- Software as a service – Provides fully designed apps via the internet without requiring any setup. You just log in and start working because the provider takes care of the security, upgrades, and all the technical aspects. Used to collaborate, manage and analyze. For example.

(26) Version Control System : The project change management software that enables developers to compare updates, save versions and reverse as needed.

- Preferably used due to the fact that it is free and has open source distributed version control system that can quickly and effectively manage projects of all sizes.

- One way to think of a 'branch' is as reference to a commit. The collection of commits made up of the commit being pointed to and all of its descendants.

A 'commit' is a code version that has been preserved. Commitments are organized in a tree with one/more ancestor commitments for every commit (apart from the first).

(27) A Jupyter Notebook is what allows data scientists to create and share papers with live code, equations and other multimedia elements using an open source online application.

(21) REST API – An architectural style that is frequently regarded as the ~~for~~ norm for creating the networked applications that drive the internet. It offers a set of guidelines and limitations that, when adhered to, provide web services that are straightforward, scalable and simple to intergrate.

- REST API – Representational State transfer application programming interface.

- GET – A method for obtaining information from a sever
  POST – Sends information to a sever in order to generate new resources.
  PUT – Used to replace or update a resource that already exists
  DELETE – This command eliminates a resource from a sever.

(22) JavaScript Object Notation (JSON) – Is a common text-based format that uses JavaScript object syntax to describe structured data. In online applications, it is frequently utilized for data transmission.

- Highly preffered for their user-friendliness and adaptability with the majority of programming languages.

- Example – { Tshegofatso Molefi, 24, Botswana }

(23) Prompt Engineering – The act of crafting, honing and optimizing inputs to motivate generative AI systems to produce precise, superior results.

- Prompt engineering matters because tools like chat GPT or claude don't think the way humans do, they respond based entirely on how your request is framed. The quality of the output is

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(29) SECTION D : DATA ETHICS, GOVERNANCE AND CAREERS

Data Privacy – The ~~protection~~ capacity for people to choose who may access their personal information and the security of personal data from unauthorized parties.

- IMPORTANCE : Ensures that private data is protected, encourages creativity, builds trust and complies with legal obligations

- The General Data Protection Regulation is an extensive set of privacy laws enacted by the EU to safeguard the gathering, use and transfer of personal data belonging to EU citizens. Applies to countries within the European

Union, or European Economic Area

(20) Data Ethics – The moral duties and guiding principles that govern an organization's collection, sharing and use of data, particularly personal data.

① CATFISHING – This is done especially on dating apps or social media platforms where people portray themselves as different people and use these accounts to perform dangerous crimes

② JOB LOSSES – AI has replaced humans in the industry and took over, especially in instances of robotics. An increase in job losses is a problem for the economy

③ DISPUTES IN INTELLECTUAL PROPERTY – People can now use AI wrongly to copy and target off work that is not their and did not create. This means the actual creator may loose on income because of this

(21) Data governance is all you do to guarantee that data is safe, confidential, accurate, accessible and useful. It encompasses the steps individuals must take, the procedures they must adhere to, and the technology that helps them along the data life cycle

- Vision strategy: Describe long-term objectives for data governance such as data monetization, compliance and trust. Link the data governance role to strategic and commercial objectives
- Data management: Assign accountability for upholding compliance and data quality
- Data quality management: Establish procedures for validations, cleaning and profiling
- Data management process: Describe the creation, storage, access and sharing of data

- Commonly used with Python.

- Popular for its tools for experimentation, data analysis and documentation.

(28) Python is a high-level object orientated programming language with simple syntax that is easy to understand

- Commonly used in the programming industry for its simple learning curve; its well-known simplicity in syntax and readability. A vast majority of other reasons may be for its enormous library collection, extensive data analytics tools.

- Pandas - which stands for Python data analysis, is an open-source data manipulation tool that is versatile, user-friendly, and has higher-level features including data structures and operations.

2. Matplotlib - A variety of data visualizations, including static, animated and interactive ones, maybe produced with the aid of the Matplotlib data visualization package.

3. Seaborn - Another python data visualization package that expands upon Matplotlib and provides high level utilities to simplify complicated data visualization.

4. SciPy - Short for Scientific Python is a library for advanced statistical calculations that maybe used to manipulate data in a variety of contexts.

~~(29) Data Privacy~~

Tableau provides a wide variety of visuals as well. Actually, it is the most well known feature. However, much as with their dashboards, the authors chosen query pathways will restrict your ability to capture.

- Helps businesses do forecasting.

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SECTION E : PERSONAL REFLECTION

35 This assignment challenged me in the sense that I almost gave up midway. It taught me patience

36 I found data governance complex because laws in different countries differs which makes it difficult to search and understand information.

37 I see myself earning lots of money online especially on ~~coffee~~ cobra.

- Technology infrastructure : Make use of resources such as security systems, metadata repositories and data catalogues

(30) AI bias, also called machine learning bias or algorithm bias, refers to the occurance of biased findings brought forth by human prejudices that corrupt the initial training data or AI algorithm, producing distorted outputs and sometimes dangerous effects.

Examples : ~~High~~ Hiring and Recruitment - Regardless of a good CV, AI algorithms may look for only certain data depending on how they have been trained which may disadvantage individuals and also promotes unfairness in the workplace. This may lead to well deserving people missing out on job opportunities

(32)

Data Analyst	Scientist	Engineer	AI Engineer
• Data collecting and cleaning • Data interpretation	• Building an ML model • Statistical analysis	• Data wrangling • Collaboration	• Data Pipelines and infrastructure

(33) Business Intelligence - helps businesses make more data driven decisions by combining business analytics, data mining, data visualization, data tools and infrastructure, and best practices

• Power BI provides a wide variety of visuals. Even though these visualizations are interactive, pre-selected query pathways will restrict your ability to filter and explore